

2026

# NET731 – Week 5

**RBAC, NSG & Storage Security | 03 – 09 March 2026**

TRISTAN PIEK

NET731 | Week 5 of 10

## DECLARATION OF AUTHENTICITY AND RULES AND REGULATIONS – THEORY

I \_\_\_\_\_ Tristan Piek \_\_\_\_\_  
(FULL NAME)

hereby declare that the contents of this assessment

\_\_\_\_\_ NET 731 FA3 Week 5 \_\_\_\_\_

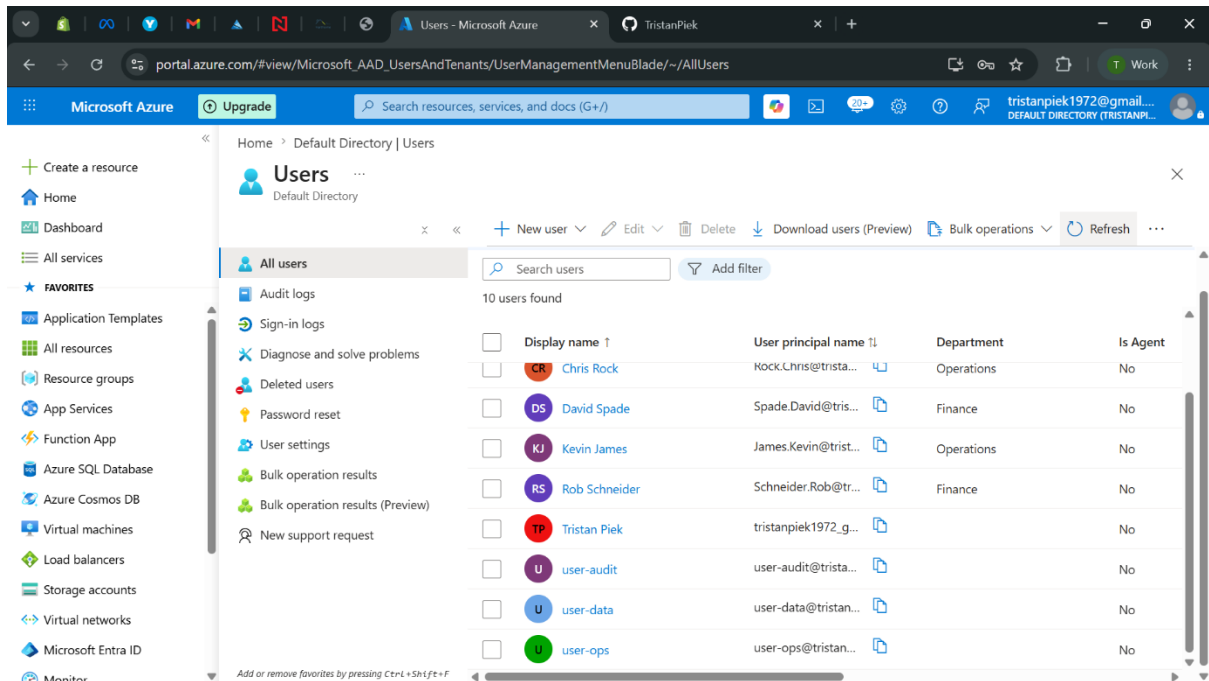
is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any **AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.**

Signature:  \_\_\_\_\_

Date: \_\_\_\_\_ 10/05/2026 \_\_\_\_\_

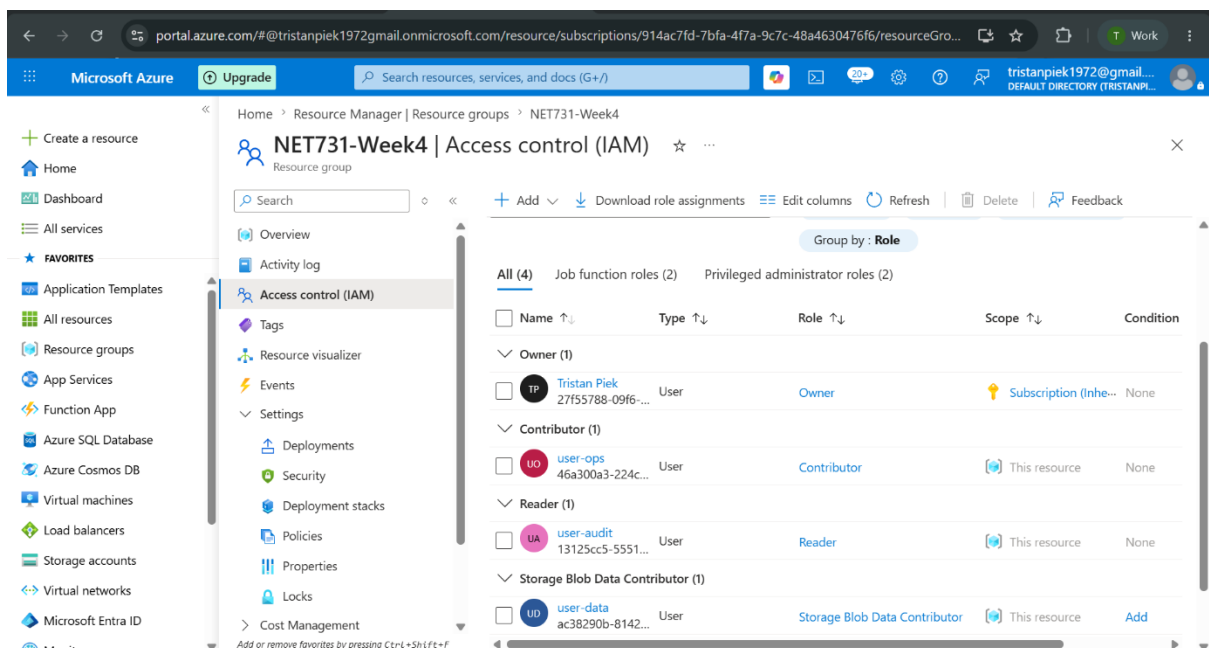
## Section 1 – Role-Based Access Control (RBAC)

### User Accounts Created



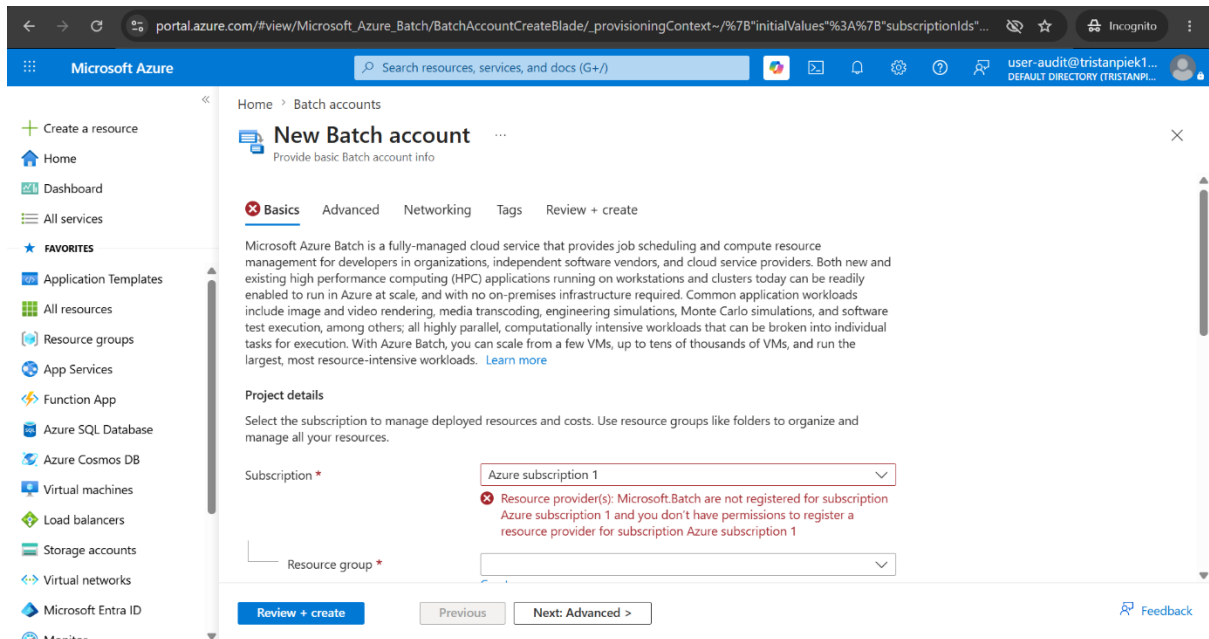
Creating the three user accounts for the practical in Microsoft Entra ID.

### Contributor Role Taxonomy



Contributor role assigned to the user-ops account for the resource group

## Verification of Reader Access Limits



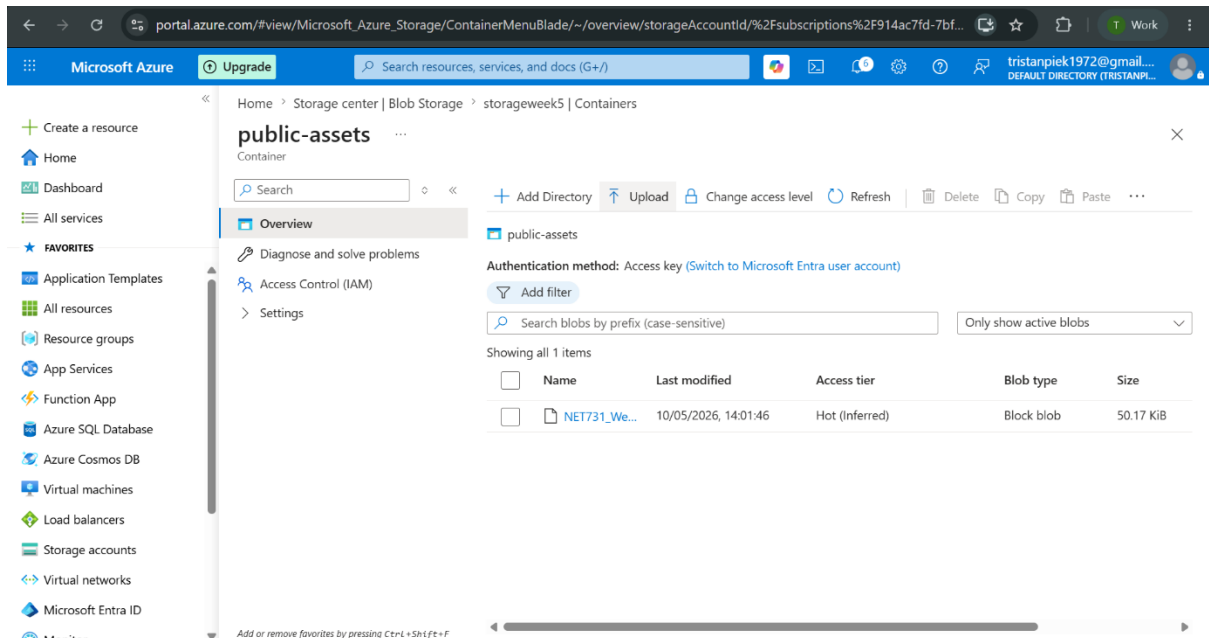
The user-audit account was unable to modify resources because of limited Reader permissions.

## Scope Clarification

Assigning roles at the subscription scope gives permissions to all resources in the subscription which breaks the least-privilege principle. Resource Group Scope Access is limited to only resources in that resource group. If Contributor was provided at subscription scope the user could change resources not related to their work throughout the environment.

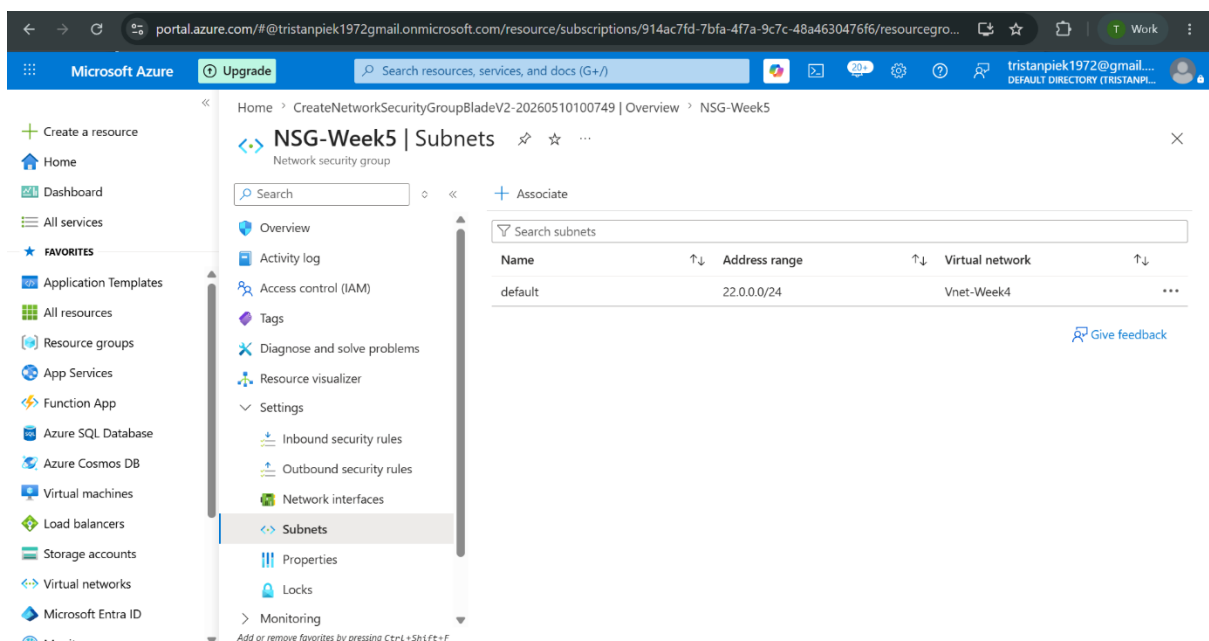
## Section 2 — Network Security Group (NSG)

### NSG Overview



The Network Security Group was successfully created in the resource group for week 5.

### NSG Subnet Association



NSG was associated to the subnet for enforcement of security rules on network traffic.

## NSG Inbound Rules

Home > CreateNetworkSecurityGroupBladeV2-20260510100749 | Overview > NSG-Week5

### NSG-Week5 | Inbound security rules

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

| Priority | Name                   | Port | Protocol | Source            |
|----------|------------------------|------|----------|-------------------|
| 100      | Allow-RDP              | 3389 | TCP      | 169.0.58.39       |
| 110      | Allow-HTTP             | 80   | TCP      | Any               |
| 120      | Allow-HTTPS            | 443  | TCP      | Any               |
| 1500     | Deny-All-Inbound       | Any  | Any      | Any               |
| 65000    | AllowVnetInBound       | Any  | Any      | VirtualNetwork    |
| 65001    | AllowAzureLoadBalan... | Any  | Any      | AzureLoadBalancer |
| 65500    | DenyAllInBound         | Any  | Any      | Any               |

Allowed remote and web inbound traffic; blocked all other inbound traffic via inbound NSG rules

## Outbound NSG Rules

Home > CreateNetworkSecurityGroupBladeV2-20260510100749 | Overview > NSG-Week5

### NSG-Week5 | Outbound security rules

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

| Priority | Name                      | Port | Protocol | Source         |
|----------|---------------------------|------|----------|----------------|
| 100      | Deny-Internet-Outbound... | Any  | Any      | Any            |
| 65000    | AllowVnetOutBound         | Any  | Any      | VirtualNetwork |
| 65001    | AllowInternetOutBound     | Any  | Any      | Any            |
| 65500    | DenyAllOutBound           | Any  | Any      | Any            |

Added outbound rule to block internet traffic going out of the virtual machines.

## Port Selection Explanation Rule for remote management

The remote management rule was created with port 3389 and TCP protocol as 3389 is the default port that Remote Desktop Protocol (RDP) uses for windows remote administration. Limiting the source to “My IP” allows only the administrator’s device to manage the VM remotely, thus improving security.

## Web Application Access Rule

Web application access was selected to use port 80 with the TCP protocol because it is the default port for HTTP web traffic. This allows users to access the hosted web application via a web browser.

## Why TCP Was Chosen

Why TCP? RDP and HTTP both need reliable, connection-oriented communication to allow the proper transfer of data between systems.

## What is Explicit Deny Rule?

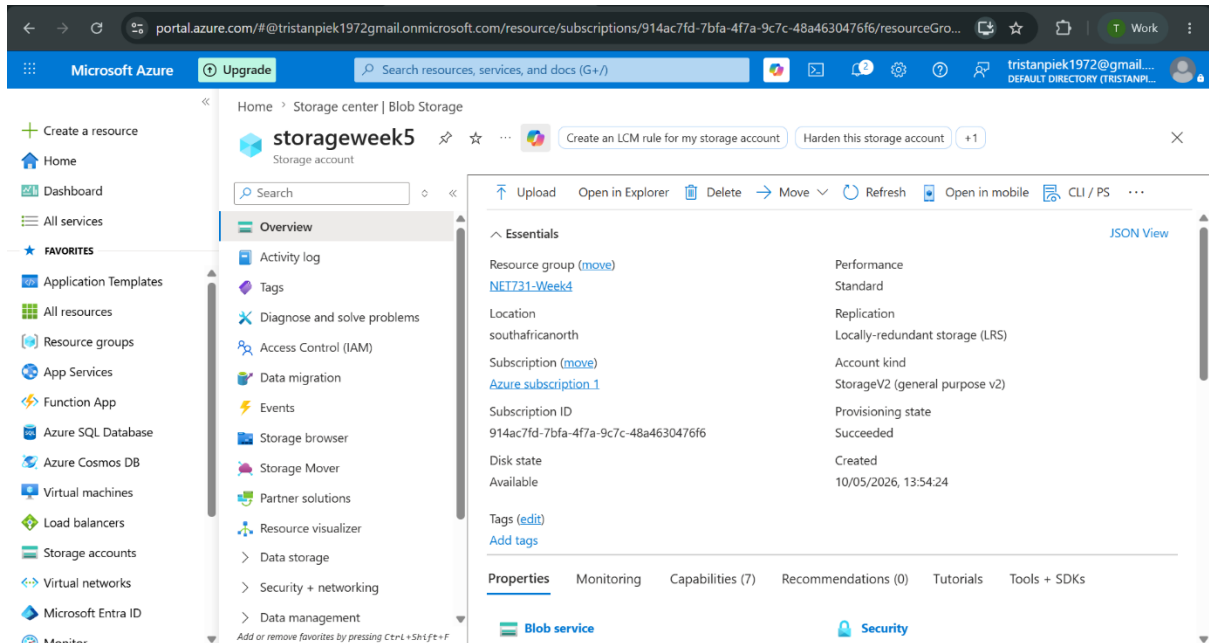
Azure NSGs have a default DenyAllInbound rule with priority 65500. If you want to ensure that your organisational security policies are enforced explicitly before the default rules, you create a custom deny rule with a lower priority.

## Rule Priority Description

NSG rules are evaluated from the lowest to the highest priority number. Blocking rules need to be applied after evaluating valid traffic, so deny rules should have higher priority values than allow rules.

## Section 3 — Storage Account and Containers

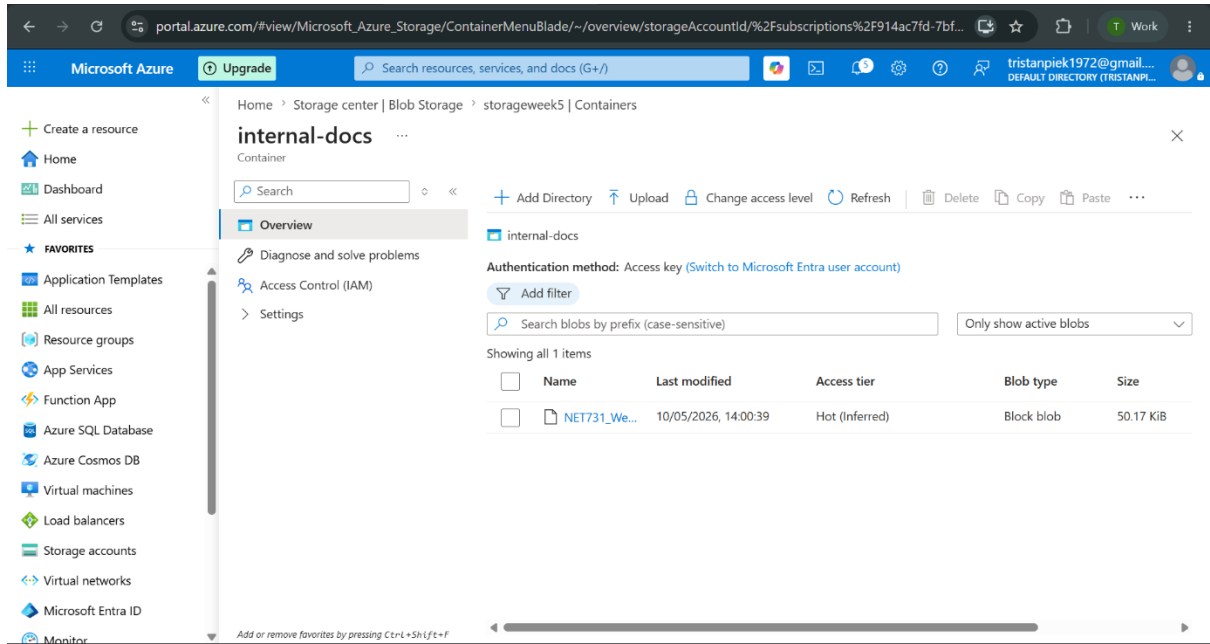
### Storage Account Overview



Storage account created with LRS replication and Hot access tier



## File verification upload



The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'internal-docs' container overview. It includes a search bar, a list of actions (Add Directory, Upload, Change access level, Refresh, Delete, Copy, Paste), and a table of blobs. The table shows one item: 'NET731\_We...' with a size of 50.17 KiB.

Home > Storage center | Blob Storage > storageweek5 | Containers

### internal-docs

Container

Search

+ Add Directory ↑ Upload Change access level Refresh Delete Copy Paste ...

internal-docs

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

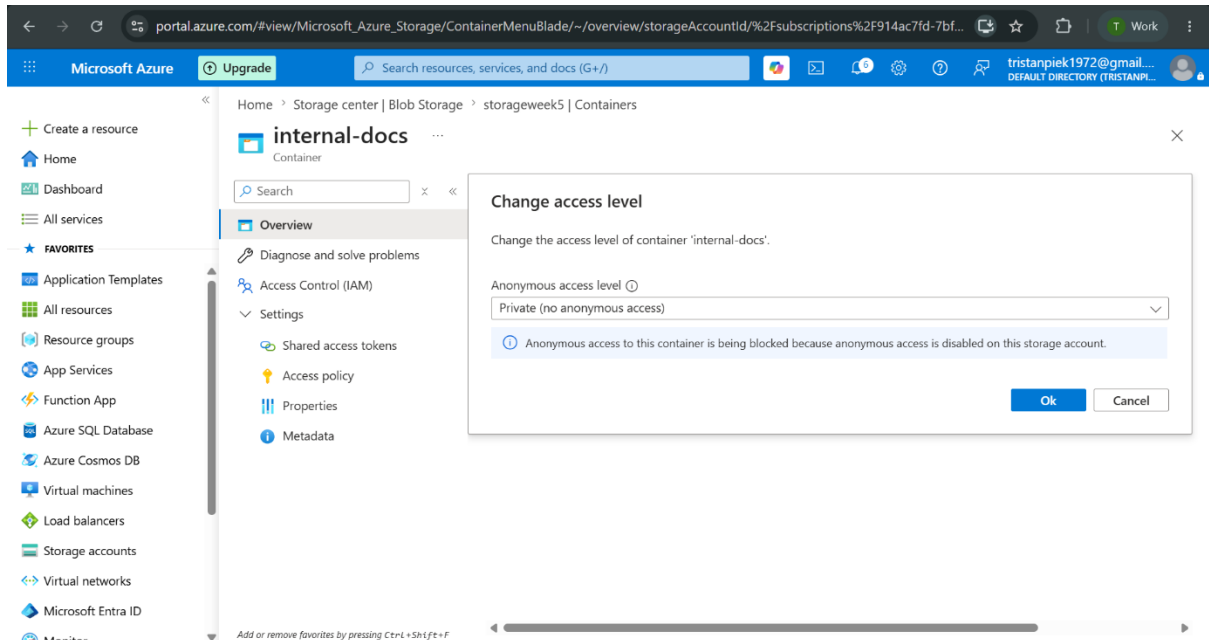
| <input type="checkbox"/> | Name         | Last modified        | Access tier    | Blob type  | Size      |
|--------------------------|--------------|----------------------|----------------|------------|-----------|
| <input type="checkbox"/> | NET731_We... | 10/05/2026, 14:00:39 | Hot (Inferred) | Block blob | 50.17 KiB |

Add or remove favorites by pressing Ctrl+Shift+F

Sample file uploaded successfully to the storage container

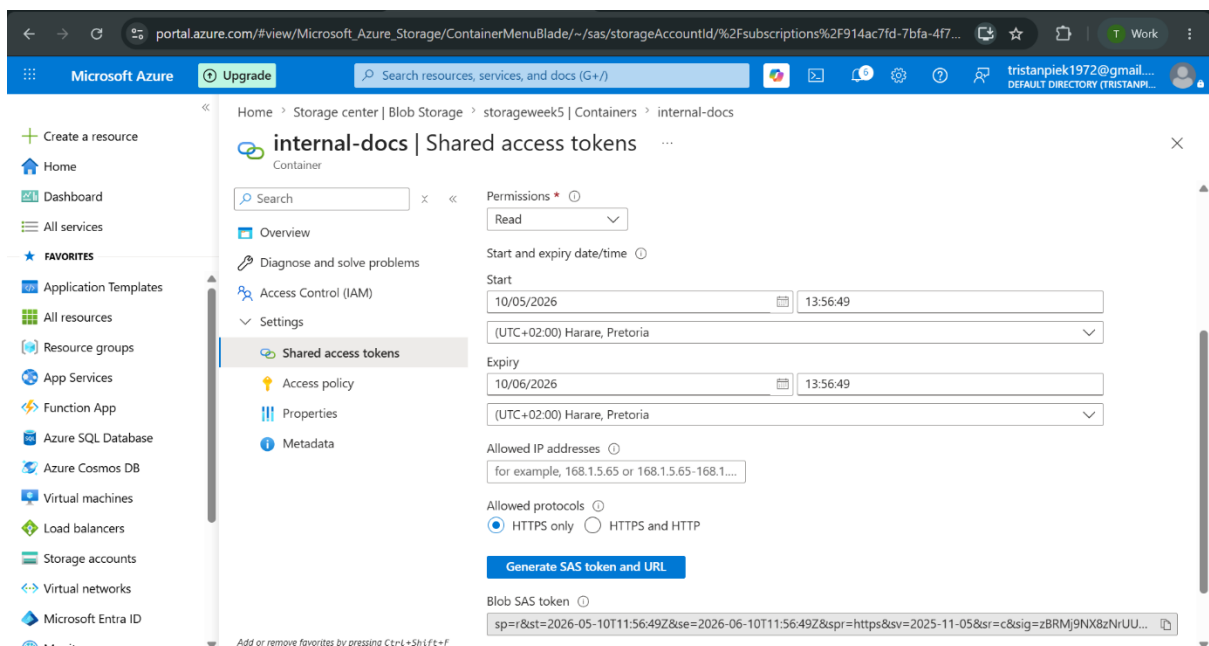
## Section 4 — Storage access permissions

### Private container access



The storage container was configured with private access to block anonymous access

### SAS Token Settings



SAS token in read-only, HTTPS-only mode.

## SAS Security Explained

SAS tokens allow you to get straight to your storage resources without needing to authenticate with Azure AD. Shared SAS tokens, if made public or shared insecurely, allow unauthorised users to access storage resources until the token expiration. If the storage account keys are exposed publicly, they should be rotated immediately, which will invalidate all active SAS tokens.